

Chapter 1

Towards a typology of tonal systems

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1 Definition of tonal language

In this project, we aim at a comprehensive typology of tonal systems. It is thus important to delineate the set of all tonal languages of the world in order to assess the variability that we should strive to represent. A lot of progress has been made in over a century since [Beach \(1923\)](#) explicitly asked “What is a tonal language?”, but the answer to this question still remains far from obvious.

Regarding the definition of a tonal language, we subscribe to the morphological view of the nature of tone first suggested by [Welmers \(1959, 1973: 80\)](#):¹

A tone language is a language in which both pitch phonemes and segmental phonemes enter into the composition of at least some morphemes.

[Hyman \(2001: 1367, 2006: 229\)](#) adapted this definition to give it its current canonical form, whereby a tonal language is one

“...in which an indication of pitch enters into the lexical realization of at least some morphemes”.

¹Cf. an earlier definition of tone as “lexically significant, contrastive, but relative pitch on each syllable” by [Pike \(1948: 3\)](#), which is obviously too narrow. In fact, by the time of Pike, languages where pitch is only distinctive in a small share of syllables, like Lithuanian or Japanese, have long featured in the discussion of tone, cf. ([Beach 1923: 79](#)) and references therein.

However, there are two grey zones where this definition may be difficult to apply.

The first one concerns the relations between tone and stress. With respect to the latter, we proceed from Hyman's (Hyman 2006: 231) definition:

A language with stress accent is one in which there is an indication of word-level metrical structure meeting the following two central criteria:

- a. OBLIGATORINESS: every lexical word has at least one syllable marked for the highest degree of metrical prominence (primary stress);
- b. CULMINATIVITY: every lexical word has at most one syllable marked for the highest degree of metrical prominence.

While Hyman's definition of tone is substantial, his definition of stress is structural. As long as a suprasegmental phenomenon satisfies all three structural conditions specified by the definition (applies to a syllable rather than a mora; is obligatory within a domain that could be viewed as a prosodic word; is culminative within the same domain), it counts as stress, regardless of whether its substance is pitch, amplitude, duration, or whatever. This makes it possible for the same phenomenon to qualify as both tone (because it involves paradigmatic pitch distinction) and stress (because it has the right syntagmatic properties).

The problem here is that stress typically has several interrelated acoustic correlates and it may not be straightforward to determine whether pitch is the primary exponent of stress or a byproduct of other, more salient properties. For instance, Levi (2005) shows that while duration, amplitude, and pitch are all involved in the distinction between stressed and non-stressed syllables in Turkish, it is pitch that provides the most reliable cue. Thus, by the latter of Hyman's definition, Turkish should be included into the ranks of tonal languages. What is more, there is a large class of languages whose tonal vs. non-tonal status can only be determined on the basis of perceptual experiments.

To avoid the associated conundrums, for the purposes of this project, we simply exclude those languages where tone can also be described as stress from consideration for the time being. Hyman's definition of tonal language thus has to be supplemented:

A language is tonal if an indication of pitch enters into the lexical realization of at least some morphemes. A language is not considered as tonal if its word-level suprasegmental contrasts can be exhaustively formulated in terms of stress, even if the stress is expressed by pitch.

While this narrowing of the definition of tonal language is admittedly somewhat opportunistic, it is not without precedent. Probably the best known borderline case between tone and stress is represented by Nubi, an Arabic-based creole spoken in Kenya and Uganda, where each lexical word has exactly one H-toned syllable. Hyman (2006: 230) recognizes Nubi as tonal language, because it complies with his definition thereof. A different approach is taken by Gussenhoven (2006: 214–218). While both authors agree that Nubi presents a “pivotal” case between a tone system and a stress system, Gussenhoven argues that it does not show the properties characteristic of “typical” tone or stress languages and thus should not count as either. He proposes restrictions of both definitions in order to exclude Nubi. Although the wording is not identical and, more importantly, our reasons for narrowing down the definition are very different from Gussenhoven’s, our supplemented definition given above delineates essentially the same set of languages as does the one proposed in (Gussenhoven 2006: 215).²

Another debatable area is represented by “mixed pitch/phonation languages” and “(voice-)register languages” which are mainly spoken in Southeast Asia (although similar phenomena may also be found elsewhere), for a survey see (Kirby & Brunelle 2017).

In pitch/phonation languages, tonemes are characterized not only by pitch, but also by specific phonation settings (modal, breathy, creaky, tense, etc.). In register languages, which are sometimes hard to distinguish from pitch/phonation languages, phonologically meaningful contrasts are expressed jointly by a bundle of (quasi-)suprasegmental features: phonation type, pitch, vowel quality, intensity, vowel duration. Just like in the case of stress, for a specific language variety it may be far from obvious which of the bundled features provides the dominant cue for identification of a register.

The common practice in SEA linguistics is to consider the pitch/phonation language as tonal, and the register languages as non-tonal: in the former, the primary cue is pitch, and the latter, phonation. For lack of better alternatives, at present we opt to stick to this practice, even though it is nowadays generally acknowledged that the boundary is rather fuzzy (Kirby & Brunelle 2017).

2 Comparative analysis of tonal systems

Tone may function in very different ways in the languages of the world. Languages vary in which pitch gestures (either level or contour) are contrasted

²Note that while we do not consider Nubi and similar languages tonal, we have no problem with positing stress in them, contrary to Gussenhoven.

phonologically, which lexical or grammatical meanings are expressed with the help of tone, and how widely tone is distributed across lexicon and grammar. The ways in which tones are related to phonological segments are likewise non-uniform. Contrasting tones and melodies may map to moras, syllables or words, or extend to phrases. Tones sponsored by a morpheme may float around, appear on a different segment, sometimes far apart, and quite often one finds purely tonal morphemes for which tone is the only exponent. Lastly, tones may or may not interact with each other, as well as with stress and intonation. The presence and intensity of such interactions also varies widely.

The dimensions of variation are rich enough so that we can characterize a particular configuration of tonal phenomena in a language as a *tonal system*. The systemic look at the functioning of tone is supported by the complexity of the interaction of lexical tone with other linguistic domains. Our goal in this project is to systematically investigate the diversity and variation of tonal systems. This investigation requires, first of all, a thorough description of tonal systems of a genetically, typologically, and geographically diverse sample of languages. But most importantly, it requires a methodology that would allow one to systematically compare tonal systems to each other.

A thing to avoid along the way is the holistic approach to typology where a few distinct types of tonal systems are sought. While holistic approach seems to be generally out of fashion in typology nowadays, it may persist in certain corners. For tone, especially when it is not the central research topic, the most widely employed contrast is that between tonal languages and stress languages. This prosodic dichotomy infamously gave rise to an intermediate type of the so-called pitch-accent languages, over which much ink has been spilled in tonology (Ding 2006, Van der Hulst 2011, Hyman 2006, 2009). As is now evident, tone and stress may both be either present or absent in a language, and interact in more or less complex ways. Seeing pitch-accent languages as a kind of hybrid type in fact exaggerates contrasts between tonal and stress languages while masking internal diversity of each type. A similarly problematic dichotomy opposes tonal languages and register languages (Kirby & Brunelle 2017).

Binary classifications are sometimes evoked to characterize the internal structure of a tonal system. The most traditional is the separation of level-based and contour-based tonal systems. This dichotomy often goes hand in hand with the opposition of simple (two-way level contrast, usually high vs. low) and complex tonal systems (all the rest). As the data presented in this volume demonstrate, this way of classifying tonal systems lacks empirical support. The alternating realization of a toneme as level in some contexts and as contour in others is quite common across languages in our sample (e.g. Bambara, Tibetan, Iquito, Japanese).

In general, mixing levels and contours in the same system seems to be a norm rather than an exception. Moreover, the phonologically simplest system with a single toneme may employ a contour as a distinctive unit (Iquito, Japanese).

Hyman (2018) argues for the properties-based approach to a typology of tonal systems instead. The foundational observation for this approach is that various tone-related phenomena may (and do) vary independently of each other. Extending the list of properties to all the aspects of tone that vary across languages of the world is thus a way forward to build a typology able to adequately characterize the diversity of tonal systems. This is the approach we shall take in the present volume: to describe tonal systems of a diverse sample of languages according to a standardized set of properties. The problem to solve is a sufficiently informative set of properties that may serve as typological variables.

In some respects, the properties of tonal systems are well-trodden ground for tonology. A number of Greenbergian style universalia about the complexity of the system of tonal contrasts were formulated early on by Maddieson (1978). A few general principles and tendencies were stated in application to the tonal processes (Hyman & Schuh 1974) and, after a few decades of expansion of empirical data on tone, these principles stand quite well (Hyman 2007). A recent overview chapter on tone by Hyman & Leben (2021) covers a comprehensive list of varying features of tonal systems. We build up on these features for our set of properties for the description of a tonal system.

Yet systematic quantitative data on all the relevant properties of tonal systems across a diverse sample of languages still seem to be a missing piece in the typology of tone. In a modern typological paradigm of “what’s where why?” research questions (Bickel 2007) these data would be an answer to a *where* part. As Hyman himself admits, phonologists studying tone were not so interested in the *where* question, giving primary attention to how to analyse the system (Hyman 2018), as well as discussing logical (im)possibilities of tonal phonology in general. In that framework the empirical evidence from tonal systems of various languages serve primarily as illustrative examples. As a result, the range of possible values for each parameter of a tonal system is much better documented than their distribution across languages.

This is not to say that comprehensive quantitative data on the tonal systems does not exist. But the scope of the tonal system parameters that got into the typological databases and areal typological studies is rather limited compared to the list of relevant features known to tonology. The most accessible general and phonological databases (WALS, LAPSyD) essentially contain two parameters with respect to tonal systems: (i) the presence of tone in the language, (ii)

and the complexity of the system of tonal contrasts (in graded categories: simple/medium/complex) (Maddieson 2013, Maddieson et al. 2013). Recent typological studies that consider interactions with tone are also likely to employ similar simplistic parameters as their measurements of tone (Easterday 2019). While this is still informative, it is, admittedly, far from a set of parameters that may adequately characterize diversity and variation of tonal systems. One still needs to bring the multiple parameters of tonal systems routinely identified in the descriptive studies and extensively discussed in the tonology to the wide-ranging quantitative typological surveys.

An additional problem for the phonological typology is the multiplicity of possible analyses (Round 2023). It is not uncommon for tonal languages to be subject to multiple conflicting interpretations of the tonal inventory, often differing in the number and quality of the tonal contrasts. Since other tonal parameters are contingent upon the posited inventory of distinctive tones, the decision to select one interpretation for a given language is highly consequential for many typological variables (tonal processes, types of tonal exponence, tonotactics, etc.). In case when such choices are not made consistently, the fundamental requirement of typology to compare like with like is undermined.

In this project, we take a normalization approach to the problem of multiple conflicting analyses of a tonal system that seeks something that can be likened to a lowest common denominator of tonal systems. Rather than adopting any single model of tonal phonology, we describe every language through a small set of cross-linguistically applicable comparative concepts, grounded as far as possible in surface distribution (Section 3). On this basis we characterize tonal systems by a standardized set of parameters elicited by the project's two analytical instruments and gathered in an online database (Section 4). The parameters themselves are briefly surveyed in Section 5, and the system we use to annotate tonal phenomena in running text in Section 6.

3 Comparative concepts for the description of tonal systems

The normalization strategy outlined above rests on a deliberately small set of analytical primitives. Rather than committing the project to any one of the competing theoretical models of tonal phonology, we frame our description in terms of *comparative concepts* in the sense of Haspelmath (2010): notions defined so as to apply across languages. Comparative concepts should be defined based on some universal grounds, rather than use the language-specific categories defined

structurally with reference to other categories in the language. In morphosyntax, semantics typically serves as the primary source for comparative concepts. In phonology, articulation is arguably a similarly universal foundation for comparative concepts suitable for linguistic typology. One particular phonological tradition, that of articulatory phonology (Browman & Goldstein 1992), served as an inspiration in the development of our set of comparative concepts for the study of tone, devised in this project. The central idea of this framework is to make phonological representations a mirror image of the articulatory intentions of a speaker.

The set of concepts we use for the comparison of tonal systems are thus motivated by function and the observable articulatory phenomena. Importantly, this focus on the surface realizations implies only minimal assumptions about underlying representations. This is what makes the values elicited for our typological variables comparable in the first place, irrespective of the diverging convictions that individual linguists may hold about the phonological and morphosyntactic derivations at work in a given language.

Two analytical decisions underlie the whole system: (i) assigning a phonological status to the pitch gestures observed in a language, and (ii) relating those gestures to a meaning, i.e. attributing them to a particular form-meaning unit. Consensus on these two points is far easier to reach than agreement on a complete inventory of underlying forms together with the derivational processes that map them onto the surface. Our comparative concepts are built around precisely these two decisions.

The central concept is the *toneme*: the minimal pitch-based phonological unit that serves to distinguish lexical and/or grammatical meanings in a language. Phonologically, a toneme consists of a single pitch target (a level toneme) or a sequence of targets (a contour toneme). Its surface counterpart is the *tonal span* – the stretch of the segmental chain over which the toneme is realized. Crucially for our purposes, the tonal span is inherently scalable: it may shrink to a fraction of a syllable or extend across several syllables or even words. We therefore do not assume a one-to-one mapping of tonemes onto syllables or morae. The criteria for establishing the tonemic status of a pitch gesture, and for telling level tonemes from contour tonemes, are developed at length in the chapter on toneme identification in this volume (Maslinsky & Vydrin 2026 [this volume]).

The second concept addresses the morphological side of tone. We represent the lexical specification of a form as an *m-item*: a pairing of a tonal specification (its tonemes) with a segmental specification (its phonemes), associated with some meaning. Either component may be empty, so that purely tonal morphemes (a tonal specification with no segments) and toneless morphemes (segments with

no tonal specification) emerge as special cases of the same notation. The label “m-item” is intentionally non-committal: it presupposes no particular theory of the morpheme and is compatible with a range of morphological frameworks.

Describing a tonal system in these terms neither denies the existence of underlying forms and tonal rules nor questions their usefulness for the analysis of an individual language. Rather, such a description is a distillate of those analyses, reduced to a form in which the material becomes comparable across languages. If the reader will indulge us in a metaphor, the one we have in mind is that of pressing. The analysis of a single tonal language yields a rich, language-particular juice, — all the underlying forms and derivations the analysis posits, indispensable for describing that language, — and, alongside it, a drier residue that comes out comparable from one language to the next. Descriptive and theoretical work thrives on the juice; cross-linguistic typology lives off the residue. Our comparative concepts are designed to extract precisely that residue.

4 Instruments and database

Our methodological approach here is very much inspired by that of the St. Petersburg/Leningrad Typology Group (Nedjalkov & Litvinov 1995, Khrakovskij & Zaika 2024), particularly in what concerns use of a standardized typological questionnaire and attention to particulars of each sampled language. The activities of this research group have mainly been concerned with morphosyntax (especially verbal valency and polypredication), but its organizational model can be applied elsewhere. We believe it to be especially relevant for the field of tonology, where typological studies are still somewhat hindered by multifarious terminological and notational traditions, abundance of competing theoretical frameworks, and vulnerability of generalizations based on incomplete datasets.

The project deploys two analytical instruments, which differ in perspective. The first, and central, instrument is a standardized questionnaire for the description of a tonal system. It poses a series of questions about the phonological inventories, both segmental and tonal, about tonotactics, tonal processes, grammatical tones, and so on. A filled-in questionnaire becomes an *analytical report* on the tonal system of the language. The questionnaire thus serves a double methodological purpose. For the author of a description it is a guide that helps organize the description and make the analytical choices that are aligned with the general framework of the project. For prospective users the analytical reports are the primary source for the standardized values of the typological variables that characterize tonal systems. In this sense the questionnaire is a paradigmatic instrument:

it aims to capture the full range of tonal phenomena attested in a language, form by form.

This is complemented by a second, syntagmatic instrument — a system for annotating tonal phenomena in running text, built on the same comparative concepts of toneme, tonal span and m-item. Each analytical report is accompanied by one or more short annotated texts, and the annotation conventions are set out in a dedicated document in this volume (Gerasimov et al. 2026 [this volume]).

Both instruments feed a common database, ThoTDB (Maslinsky et al. 2024), which brings together the values extracted from the questionnaires and from the annotated texts. The database is the basis for computing quantitative typological measures, such as the tonal density index, the text frequency of individual tonemes and of other tonal phenomena, and the intensity of tonal processes.

5 Parameters of tonal systems

The questionnaire is designed to reflect the entire range of phenomena that may vary across tonal systems, and the analytical reports it generates are correspondingly rich. From these descriptions one can read off the *parameters* of tonal systems that serve as our typological variables. The underlying model is paradigmatic: each report seeks to document the full diversity of tone-related phenomena in a single language. The depth of this description far exceeds the two or three tonal variables available in WALS and LAPSyD, which sets the present project apart from earlier typological surveys that touched on tone.

The questionnaire groups the relevant phenomena into several broad domains, which we briefly summarize below. The full questionnaire is included in the present volume as a separate chapter.

5.1 Tonemic inventory

The first domain is the inventory itself: the set of tonemes of the language together with their regular allotones, whether level or contour. Alongside the tonemes proper, the inventory records the *default tonal level* — a pitch target that carries no distinctive function, and is hence not a toneme, but that is nonetheless phonologically defined on extratonal (toneless) syllables. The definition of the toneme and the diagnostics for identifying tonemes are treated in detail in the chapter on toneme identification in this volume (Maslinsky & Vydrin 2026 [this volume]).

5.2 Tonotactics

The second domain concerns tonotactics: the distribution of tonemes over segmental strings of various types. The summary measure here is the tonal density index, which relates the number of tonemes to the number of prosodic units (syllables or moras). Under this heading we also describe the surface realization of tones, in particular, whether the language has extratonal syllables or moras that fall outside any tonal span and if so, the ways in which these syllables acquire pitch.

5.3 Typology of tonal exponents

The third domain is the coordination of tones and segments within the exponents of morphemes, i.e. the typology of tonal exponence (cf. [Kaldhol 2024](#)). The questionnaire registers, among other things, tonal morphemes whose exponent combines a segment with a tonal operation realized outside that segment, and tonal morphemes that contain more than one toneme. Recast in terms of m-items, the attested types range from purely tonal morphemes with no segmental co-exponent, through segmental morphemes with no tonal co-exponent, to morphemes that combine both. The last type subdivides further, according to whether the toneme is inherent to the segmental exponent — that is, lexically aligned to it, and thus carrying an alignment point — non-inherent (with no fixed alignment), or both at once.

5.4 Typology of tonal processes

The fourth domain is the typology of tonal processes. We define a tonal process through surface tonal alternations in the realizations of one and the same morpheme across contexts. By focusing on surface alternations, we can avoid the dependence of our data on the particularities of the derivational analysis pertinent to the more traditional understanding of the tonal processes or tonal rules. Our surface-driven distributional understanding of tonal processes wrings out most of the juice, but that makes it an effective comparative concept. The full definition is given in the chapter on toneme identification in this volume ([Maslinsky & Vydrin 2026 \[this volume\]](#)). This domain also covers the interaction of tone with intonation.

5.5 Coverage of the lexicon by tonal exponents

The final domain measures how far tonal exponents reach into the lexicon and grammar. On the grammatical side, the parameters include the presence and the

number of grammatical tonal morphemes, the grammatical meanings they express, and the categories of words that can carry grammatical tone. On the lexical side, they include the presence of lexical tone, the lexical categories or parts of speech that lack it, and whether the distribution of tonemes is conditioned by word class.

6 The system of tonal annotation

The annotation system records, for a stretch of running text, three coordinated layers of information. A surface layer captures the surface realization of tonemes — the boundaries of tonal spans and the allotones realized within them, together with toneless segments and default tones. An underlying layer, consisting of a sequence of m-items, records the morphological affiliation of the tonemes, that is, which toneme is contributed by which morpheme. A gloss layer states the lexical and grammatical meanings expressed. Taken together, these layers add a syntagmatic dimension to the paradigmatic picture built up by the questionnaire, showing how tonal phenomena are deployed in actual discourse.

Because the annotation is labour-intensive, the texts assembled so far are short. We set a minimum threshold of 100 syllables. For most languages in the sample the annotated material runs to a few hundred words. A text of this size is an independent source only for those phenomena on which even a small sample can yield meaningful statistics. These are relatively low-level facts such as the text frequency of tonemes, the size of tonal spans, the alignment of tonal spans with prosodic units, and the overall activity of tonal processes, visible in the relation between the underlying and the surface layers.

Phenomena that are tied to particular lexicalized items or infrequent forms will not occur often enough in so short a text to support statistics: they would be too rare, or absent altogether. For the typology of such phenomena we rely instead on the analytical report and the questionnaire-based records in the database. The typology of tonal processes falls into this second class: processes may be too sparse for syntagmatic analysis on a short text, and are therefore tallied from the questionnaire rather than from the text. One practical consequence for annotation is that we need not enforce a strict toneme-by-toneme correspondence between the underlying and the surface layers.

7 Distribution of tonal languages in the world

Even with Nubi-like stress languages and voice-register languages excluded, a significant share of the world's languages are tonal. The ThoT Database lists all 7674 attested spoken languages³ from Glottolog 5.0 and indicates a tonal status for each (although, as of July 2026, almost one third of all languages are marked as “Unknown” and while this number is expected to decrease as our work progresses, it will never be realistically reduced to zero). For this purpose, the database aggregates data from existing sources, such as WALS, LAPSyD, Larry Hyman's personal database of tonal languages, and, most prominently, recent global survey of the presence of tone (Hammarström & Mannby forthcoming). The latter study relies on semi-automated mining of data from published grammars and other sources and so contains a certain amount of false attributions. Nevertheless, there is an ongoing work on manual data verification with the help of experts on specific languages, families, and areas; at present, tonal statuses of over 3000 languages in the ThoT Database have been manually checked.

Our current (as of July 2026) estimate of the total share of tonal languages of the world, as proudly displayed at the frontpage of the ThoT Database, is 42.9%. This is far below oft-cited assessments of “nearly/about half” or “up to 60–70%” (Fromkin 1978: 1, Yip 2002: 17, Gussenhoven 2004: 12, Ashby & Maidment 2005: 163), it is not too far (albeit still below) from the estimate of 46.7% given in (Maddieson 2023), but, interestingly, it is pretty close to the 41.75% rate in WALS (Maddieson 2013): out of 527 languages of the sample, 220 are recognized as tonal.

Even this figure should be treated with some care. Given the procedure for its calculation, it probably should more strictly be viewed as our current estimate of the total share of languoids awarded language status in Glottolog that have at least one dialect that has been described as tonal.

8 Volume structure

Tonal languages, spoken in all macro-areas (with the very likely exception of Australia), are many and diverse. This volume, with its thirteen language chapters can of course only present a small slice of this diversity. Our goal with this collection of papers is to introduce our parametric model and demonstrate, on the basis of a small convenience sample, its applicability to genetically, areally, and typologically diverse languages. Since we do not intend to arrive at any typological generalizations yet, there is no attempt at representative global sampling at

³With the exclusion, for various reasons, of pidgins, speech registers, and mixed languages.

this stage (although the structure of the ThoT database has been developed with this purpose in mind). The composition of this volume is obviously skewed towards West Africa and do not include any languages from the Americas (volume II, currently in preparation, will remedy this to some extent).

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